

Risk measures

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Chapter 3: Statistical inference

Plan

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- 2 Estimation of risk measures
- 3 Backtesting and numerical illustrations

- 1 Quantile estimation
 - Empirical cumulative distribution function
 - Empirical Quantile
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Definition of the empirical cdf

Let $X_1, \dots, X_n$ a sample of cdf F . The empirical cdf

$$F_n(x) = \frac{1}{n} \sum_{i=1}^n 1_{X_i \leq x}$$

is the **empirical distribution**, which gives weights $1/n$ to the X_i 's.

Properties (Elementary properties)

- ① $nF_n(x) \sim \mathcal{B}\{n, F(x)\}$;
- ② $F_n(x) \xrightarrow{a.s.} F(x)$ (LLN) and $\sup_{x \in \mathbb{R}} |F_n(x) - F(x)| \xrightarrow{p.s.} 0$ (Glivenko-Cantelli, 1933) when $n \rightarrow \infty$;
- ③ $\sqrt{n} \{F_n(x) - F(x)\} \xrightarrow{\mathcal{L}} \mathcal{N}\{0, F(x)(1 - F(x))\}$ (usual CLT);
- ④ If F continuous and $x_n \rightarrow x$ a.s. then $\sqrt{n} \{F_n(x_n) - F(x_n)\} \xrightarrow{\mathcal{L}} \mathcal{N}\{0, F(x)(1 - F(x))\}$ (stochastic equicontinuity).

Multivariate CLT

Theorem (Bivariate CLT)

$$\sqrt{n} \begin{pmatrix} F_n(x_1) - F(x_1) \\ F_n(x_2) - F(x_2) \end{pmatrix} \xrightarrow{\mathcal{L}} \mathcal{N}(0_2, \Sigma),$$

$$\Sigma = \begin{pmatrix} F(x_1) - F^2(x_1) & F(x_1 \wedge x_2) - F(x_1)F(x_2) \\ F(x_1 \wedge x_2) - F(x_1)F(x_2) & F(x_2) - F^2(x_2) \end{pmatrix}$$

Note that if $x_1 \leq x_2$ then

$$\begin{aligned} & \text{Cov} \left(\frac{1}{\sqrt{n}} \sum_{i=1}^n 1_{X_i \leq x_1}, \frac{1}{\sqrt{n}} \sum_{j=1}^n 1_{X_j \leq x_2} \right) \\ &= E 1_{X_1 \leq x_1} - E 1_{X_1 \leq x_1} E 1_{X_1 \leq x_2} \\ &= F(x_1) - F(x_1)F(x_2). \end{aligned}$$

Brownian bridge

Let $\{W(t), 0 \leq t \leq 1\}$, a **Brownian motion** ($W(0) = 0$, independent increments, continuous trajectories, $W(u) - W(s) \sim \mathcal{N}(0, |u - s|)$).

► Simulations

If $t_2 \geq t_1$, we have

$$\text{Cov}\{W(t_1), W(t_2)\} = \text{Cov}\{W(t_1), W(t_1) + W(t_2) - W(t_1)\} = t_1 \wedge t_2.$$

The process $B(t) = W(t) - tW(1)$ is called a **Brownian bridge**: a Gaussian process such that

$$B(0) = B(1) = 0, \quad EB(t) = 0$$

► Simulations

and

$$\text{Cov}\{B(t_1), B(t_2)\} = t_1 \wedge t_2 - t_1 t_2.^1$$

¹indeed

$$\text{Cov}\{W(t_1) - t_1 W(1), W(t_2) - t_2 W(1)\} = t_1 \wedge t_2 - t_1 t_2 - t_2 t_1 + t_1 t_2$$

Functional CLT

Theorem (Donsker, 1952)

In the set $\mathcal{D}((-\infty, \infty))$ of the cadlag (continues à droite, limite à gauche) functions endowed with the Skorokhod distance^a we have the weak convergence^b

$$\sqrt{n} \{F_n(x) - F(x)\} \Longrightarrow G(x) = B(F(x)),$$

where B is a Brownian bridge, hence G is a centered process with covariance:

$$\text{Cov}\{G(x_1), G(x_2)\} = F(x_1) \wedge F(x_2) - F(x_1)F(x_2).$$

^asee Billingsley, P. (1999) *Convergence of Probability Measures*.

^bfor any function $f : \mathcal{D}((-\infty, \infty)) \rightarrow \mathbb{R}$ which is continuous (in the sense of the Skorokhod distance), we have $f(\sqrt{n} \{F_n(x) - F(x)\}) \xrightarrow{\mathcal{L}} f(B(F(x)))$.

Example

The Kolmogorov-Smirnov statistic

$$D_n = \sup_{x \in \mathbb{R}} |F_n(x) - F(x)|$$

is used to test

$$H_0 : X_i \sim F, \quad \text{against} \quad H_1 : \text{the distribution of } X_i \text{ is not } F.$$

When $n \rightarrow \infty$, we have

$$\sqrt{n}D_n \xrightarrow{\mathcal{L}} \sup_{x \in \mathbb{R}} |B(F(x))| = \sup_{y \in [0,1]} |B(y)|.$$

Computation by classification

For $\alpha \in (0, 1)$,

$$\xi_\alpha = F^-(\alpha), \quad \xi_{n,\alpha} = F_n^-(\alpha) := \inf\{x : F_n(x) \geq \alpha\}.$$

Let the **order statistics**

$$X_{1,n} = \min\{X_1, \dots, X_n\} \leq X_{2,n} \leq \dots \leq X_{n,n} = \max\{X_1, \dots, X_n\}.$$

The cdf F_n is equal to $\frac{k}{n}$ on $[X_{k,n}, X_{k+1,n})$ (if this interval is not empty). Therefore $\xi_{n,\alpha} = X_{k,n}$ with $k = k_{\alpha,n}$ such that $\frac{k-1}{n} < \alpha \leq \frac{k}{n}$. We thus have

$$F_n^-(\alpha) = \sum_{k=1}^n X_{k,n} 1_{(\frac{k-1}{n}, \frac{k}{n}]}(\alpha), \quad \xi_{n,\alpha} = X_{\lceil n\alpha \rceil, n}, \quad \text{Illustration}$$

where $x \leq \lceil x \rceil < x + 1$, $x - 1 < \lfloor x \rfloor \leq x$:

$\lceil x \rceil = \min\{n \in \mathbb{Z} : n \geq x\}$ is the ceiling function,

$\lfloor x \rfloor = \max\{n \in \mathbb{Z} : n \leq x\}$ is the floor function.

Computation by minimization

Apparently more involved, but provides interesting generalizations.
We have seen that

► Illustration

$$\xi_{n,\alpha} = \inf\{\arg \min_z \varphi_{n,\alpha}(z)\},$$

where

$$\varphi_{n,\alpha}(z) = E_{F_n} \rho_\alpha(X - z) = \frac{1}{n} \sum_{i=1}^n \rho_\alpha(X_i - z)$$

and $\rho_\alpha(u) = u(\alpha - 1_{u < 0})$.

► Proof

CLT for a triangular array

Lindeberg's CLT

Suppose that, $\forall n > 0$, the real r.v. $Y_{n1}, Y_{n2}, \dots, Y_{nn}$ are independent, $EY_{ni} = 0$, $\text{Var}Y_{ni} = \sigma_{ni}^2$ and $s_n^2 = \sum_{i=1}^n \sigma_{ni}^2 > 0$ for all n .

Under [Lindeberg's condition](#)

► Interpretation

$$\forall \varepsilon > 0, \quad \frac{1}{s_n^2} \sum_{i=1}^n EY_{ni}^2 1_{\{|Y_{ni}| \geq \varepsilon s_n\}} \rightarrow 0 \text{ when } n \rightarrow \infty,$$

we have

$$\frac{1}{s_n} \sum_{i=1}^n Y_{ni} \xrightarrow{\mathcal{L}} \mathcal{N}(0, 1).$$

► Implies the usual CLT

Asymptotic distribution of the empirical quantile

Theorem

If $\alpha \in (0, 1)$ and if $X_1, \dots, X_n$ is an iid sequence with density f such that $f(\xi_\alpha) > 0$ and f is continuous at ξ_α , then

$$\sqrt{n}(\xi_{n,\alpha} - \xi_\alpha) \xrightarrow{\mathcal{L}} \mathcal{N}\left(0, \frac{\alpha(1-\alpha)}{f^2(\xi_\alpha)}\right).$$

Proof: for all $x \in \mathbb{R}$, we have

$$P\left\{\sqrt{n}(\xi_{n,\alpha} - \xi_\alpha) \geq x\right\} = P\left(\xi_{n,\alpha} \geq \xi_\alpha + \frac{x}{\sqrt{n}}\right).$$

Noting that $\{\xi_{n,\alpha} \geq z\}$ iff $\frac{\partial \varphi_{n,\alpha}(z)}{\partial z^-} = -\alpha + \frac{1}{n} \sum_{i=1}^n 1_{X_i < z} < 0$, we have

$$P\left\{\sqrt{n}(\xi_{n,\alpha} - \xi_\alpha) \geq x\right\} = P\left(\frac{1}{n} \sum_{i=1}^n 1_{X_i < \xi_\alpha + \frac{x}{\sqrt{n}}} < \alpha\right).$$

Proof (continued)

$$\begin{aligned}P\left\{\sqrt{n}(\xi_{n,\alpha} - \xi_\alpha) \geq x\right\} &= P\left(\frac{1}{n} \sum_{i=1}^n 1_{X_i < \xi_\alpha + \frac{x}{\sqrt{n}}} < \alpha\right) \\&= P\left(Z_n < \frac{n(\alpha - p_n)}{\sqrt{np_n(1 - p_n)}}\right),\end{aligned}$$

where

$$Z_n = \sum_{i=1}^n Y_{ni}, \quad Y_{ni} = \frac{1_{X_i < \xi_\alpha + \frac{x}{\sqrt{n}}} - p_n}{\sqrt{np_n(1 - p_n)}}, \quad p_n = P\left(X_1 < \xi_\alpha + \frac{x}{\sqrt{n}}\right).$$

We have

$$\sqrt{n}(\alpha - p_n) = \sqrt{n} \int_{\xi_\alpha + \frac{x}{\sqrt{n}}}^{\xi_\alpha} f(y) dy \rightarrow -xf(\xi_\alpha)$$

and $p_n \rightarrow \alpha$ as $n \rightarrow \infty$. To conclude, we use Lindeberg's CLT

$$Z_n \xrightarrow{\mathcal{L}} \mathcal{N}(0, 1), \quad \text{as } n \rightarrow \infty.$$

Asymptotic distribution of the empirical quantile

Theorem

If $\alpha \in (0, 1)$ and if $X_1, \dots, X_n$ iid with density f such that $f(\xi_\alpha) > 0$ and f continuous at ξ_α , then

$$\sqrt{n} (\xi_{n,\alpha} - \xi_\alpha) \xrightarrow{\mathcal{L}} \mathcal{N} \left(0, \frac{\alpha(1-\alpha)}{f^2(\xi_\alpha)} \right).$$

Alternative proof: Under the assumption of the theorem, it can be shown that $\xi_{n,\alpha} \rightarrow \xi_\alpha$ a.s. as $n \rightarrow \infty$. By stochastic equicontinuity and CLT, the empirique cdf F_n satisfies

$$\sqrt{n} \{F_n(\xi_{n,\alpha}) - F(\xi_{n,\alpha})\} \xrightarrow{\mathcal{L}} \mathcal{N} \{0, \alpha(1-\alpha)\}.$$

Continuation of the alternative proof

The empirical cdf has jumps of height $1/n$ (the probability of equal X_i 's is null), thus

$$|F_n(\xi_{n,\alpha}) - \alpha| \leq \frac{1}{n}.$$

► Illustration

It follows that $\sqrt{n} \{\alpha - F(\xi_{n,\alpha})\} \xrightarrow{\mathcal{L}} \mathcal{N}\{0, \alpha(1 - \alpha)\}$. Deriving the two terms of $F(F^{-1}(y)) = y$ we obtain

$$\frac{\partial F^{-1}(\alpha)}{\partial y} = \frac{1}{f(\xi_\alpha)}.$$

The delta method² yields

$$\sqrt{n} \{F^{-1}(\alpha) - \xi_{n,\alpha}\} \xrightarrow{\mathcal{L}} \mathcal{N} \left\{ 0, \left(\frac{1}{f(\xi_\alpha)} \right)^2 \alpha(1 - \alpha) \right\}.$$

²If $g'(\theta) \neq 0$ and $\sqrt{n}(X_n - \theta) \xrightarrow{\mathcal{L}} \mathcal{N}(0, \sigma^2)$ then
 $\sqrt{n}(g(X_n) - g(\theta)) \xrightarrow{\mathcal{L}} \mathcal{N}(0, (g'(\theta))^2 \sigma^2)$

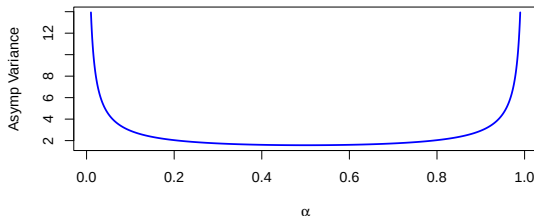
Interpretation of the asymptotic variance

$$\text{var}_{as} \left\{ \sqrt{n} (\xi_{n,\alpha} - \xi_\alpha) \right\} = \frac{\alpha(1-\alpha)}{f^2(\xi_\alpha)}.$$

The numerator is small for α 's corresponding to the tails of the distribution, but this effect is balanced by the denominator.

In general, the effect of the denominator dominates, the accuracy becoming very small in zones with a small density.

Figure: Asymptotic variance when f is $\mathcal{N}(0, 1)$.



Multivariate extension: vector of quantiles

Let $\xi_n = (\xi_{n,\alpha_1}, \dots, \xi_{n,\alpha_p})'$ and $\xi = (\xi_{\alpha_1}, \dots, \xi_{\alpha_p})'$ vectors of p empirical and theoretical quantiles.

Properties

Let $X_1, \dots, X_n$ a sample of iid variables with density f . Then, if $\alpha_i \in (0, 1)$ and $f(\xi_{\alpha_i}) > 0$ for $i = 1, \dots, p$,

$$\sqrt{n}(\xi_n - \xi) \xrightarrow{d} \mathcal{N}(0, \Omega), \quad \Omega = (\omega_{ij}), \quad \omega_{ij} = \frac{\alpha_i \wedge \alpha_j - \alpha_i \alpha_j}{f(\xi_{\alpha_i}) f(\xi_{\alpha_j})}.$$

Remark: Ω can be estimated by

$$\hat{\Omega} = (\hat{\omega}_{ij}), \quad \hat{\omega}_{ij} = \frac{\alpha_i \wedge \alpha_j - \alpha_i \alpha_j}{\hat{f}(\xi_{n,\alpha_i}) \hat{f}(\xi_{n,\alpha_j})}$$

where $\hat{f}$ is a non parametric estimator (for instance obtained by the kernel method) of the density f .

An extension to linear processes

Properties (Wu, 2005)

If $X_t = \sum_{i=0}^{\infty} a_i \epsilon_{t-i}$ where (ϵ_t) is a sequence of iid variables with finite variance and density f_{ϵ} and $\sum_{i=0}^{\infty} |a_i| < \infty$, if $\sup_x (f_{\epsilon}(x) + |f'_{\epsilon}(x)|) < \infty$, if for $0 < \alpha_0 < \alpha_1 < 1$,

$$\sup_x |f''_{\epsilon}(x)| < \infty, \quad \inf_{\alpha_0 < \alpha < \alpha_1} f(\xi_{\alpha}) > 0,$$

then

$$\sqrt{n}f(\xi_{\alpha})(\xi_{n,\alpha} - \xi_{\alpha}) \Rightarrow B(\alpha), \quad \alpha \in (\alpha_0, \alpha_1).$$

²See Wu, 2005 "On the Bahadur representation of sample quantiles for dependent sequences" Annals of Statistics

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Nonparametric estimation of VaR

One can estimate $\text{VaR}_{n,h}(\alpha) = F_{L_{n,n+h}}^-(1 - \alpha)$ by the empirical quantile of the past losses

$$\{L_{t,t+h}, \quad t = n - h - K + 1, \dots, n - h\}.$$

If, for instance, $K = 1000$ and $\alpha = 5\%$, the estimate of $\text{VaR}_{n,h}(\alpha)$ is the 50-th worst loss observed among the last 1000 losses.

Advantages: simple, available at any horizon h , can be extended to other risk measures, model free (nonparametric).

Drawbacks: one cannot use the results of the iid case because the losses are generally not independent; provides an estimate of the marginal VaR, not of the conditional one; does not account for multivariate relations between the risk factors; not very accurate for α small.

Estimation of a DRM

$$r(F, G) = \int_0^1 F^{-1}(1-u) dG(u),$$

where G cdf on $[0, 1]$, F cdf with strictly positive density over $\mathbb{R}$. The quantile $F^{-1}(u)$ pour $u \in (0, 1)$ can be estimated by

$$F_n^-(u) = \inf\{x : F_n(x) \geq u\} = \sum_{i=1}^n X_{i,n} 1_{\left] \frac{i-1}{n}, \frac{i}{n} \right]}(u)$$

and $r(F, G)$ by

$$\begin{aligned} r(F_n, G) &= \int_0^1 F_n^-(1-u) dG(u) = \sum_{i=1}^n X_{i,n} \int_0^1 1_{\left] \frac{i-1}{n}, \frac{i}{n} \right]}(1-u) dG(u) \\ &= \sum_{i=1}^n X_{i,n} \left\{ G\left(\frac{n-i}{n}\right) - G\left(\frac{n-i+1}{n}\right) \right\} = \sum_{i=0}^{n-1} X_{n-i,n} \left\{ G\left(\frac{i+1}{n}\right) - G\left(\frac{i}{n}\right) \right\} \end{aligned}$$

which is a weighted sum of order statistics, with more weights for the large values when G is concave.

Asymptotic distribution

According to Donsker, under the assumptions of Wu (2005),

$$\begin{aligned}\sqrt{n} \{r(F_n, G) - r(F, G)\} &= \int_0^1 \sqrt{n} \{F_n^-(1-u) - F^{-1}(1-u)\} dG(u) \\ &\xrightarrow{\mathcal{L}} \int_0^1 \frac{B(1-u)}{f(\xi_{1-u})} dG(u)\end{aligned}$$

with B a Brownian bridge.

The limiting distribution is centered, and its variance is given by

$$\begin{aligned}\text{Var}_{as}[\sqrt{n}\{r(F_n; G) - r(F; G)\}] \\ = \int_0^1 \int_0^1 \frac{u_1 \wedge u_2 - u_1 u_2}{f(\xi_{1-u_1}) f(\xi_{1-u_2})} dG(u_1) dG(u_2).\end{aligned}$$

Case of the ES

$$\text{ES}(\alpha) = \frac{1}{\alpha} \int_0^\alpha F^{-1}(1-u) du$$

is estimated by

$$\text{ES}_n(\alpha) = \sum_{i=0}^{n-1} X_{n-i,n} \left\{ G\left(\frac{i+1}{n}\right) - G\left(\frac{i}{n}\right) \right\}$$

where $G(u) = \frac{u}{\alpha} 1_{[0,\alpha]}(u) + 1_{] \alpha, 1]}(u)$. We have

$$G\left(\frac{i+1}{n}\right) - G\left(\frac{i}{n}\right) = \begin{cases} \frac{1}{n\alpha} & \text{If } \frac{i+1}{n} \leq \alpha \text{ (i.e. } i \leq \lfloor n\alpha \rfloor - 1) \\ 1 - \frac{i}{n\alpha} & \text{If } \frac{i}{n} \leq \alpha < \frac{i+1}{n} \text{ (i.e. } i = \lfloor n\alpha \rfloor) \\ 0 & \text{If } \alpha \leq \frac{i}{n} \end{cases}$$

Hence

$$\text{ES}_n(\alpha) = \frac{1}{n\alpha} \sum_{i=0}^{\lfloor n\alpha \rfloor - 1} X_{n-i,n} + \left(1 - \frac{\lfloor n\alpha \rfloor}{n\alpha}\right) X_{n-\lfloor n\alpha \rfloor,n}.$$

General parametric dynamic model

Consider horizon $h = 1$ and let $L_t = L_{t,t+1}$.

Commonly used models are of the form

$$L_t = m_t(\theta) + \sigma_t(\theta)\eta_t, \quad (\eta_t) \text{ iid}(0, 1), \quad \eta_t \sim F$$

where θ is parameter, $m_t(\theta)$ and $\sigma_t(\theta)$ are measurable functions of the past observations $L_{t-1}, L_{t-2}, \dots$. Assuming η_t independent from $\{L_{t-u}, u > 0\}$, we have

$$E(L_t \mid L_{t-i}, i \geq 1) = m_t(\theta) + \sigma_t(\theta)E(\eta_t \mid L_{t-i}, i \geq 1) = m_t(\theta)$$

and

$$\text{Var}(L_t \mid L_{t-i}, i \geq 1) = \sigma_t^2(\theta).$$

Standard dynamic models

$$L_t = m_t(\theta) + \sigma_t(\theta)\eta_t$$

A standard model for the conditional mean $m_t(\theta)$ is an ARMA, for instance an AR(1)

$$m_t(\theta) = a_0 + a_1 L_{t-1}.$$

A standard model for the "volatility" $\sigma_t(\theta)$ is of the ARCH (AutoRegressive Conditionally Heteroskedastic) type, for instance a GARCH(1,1)

$$\sigma_t^2(\theta) = \omega + \alpha \epsilon_{t-1}^2 + \beta \sigma_{t-1}^2(\theta), \quad \epsilon_t = L_t - m_t(\theta) = \sigma_t(\theta)\eta_t.$$

Usual constraints on $\theta = (a_0, a_1, \omega, \alpha, \beta)$ are $|a_1| < 1$ (stationarity), $\omega > 0, \alpha > 0, \beta \in (0, 1)$ ($\sigma_t^2(\theta) > 0$) and sometimes $\alpha + \beta < 1$ (stationarity and existence of a 2nd order moment).

Estimation of θ in $L_t = m_t(\theta) + \sigma_t(\theta)\eta_t$

The observations are $L_1, \dots, L_n$. The parameter can be estimated by Gaussian **quasi-maximum likelihood** (QML):

$$\hat{\theta} = \arg \min_{\theta \in \Theta} \sum_{t=1}^n \left\{ \log \tilde{\sigma}_t^2(\theta) + \frac{(L_t - \tilde{m}_t(\theta))^2}{\tilde{\sigma}_t^2(\theta)} \right\} \quad \text{► Intuition}$$

where Θ is the parameter space (taking into account, in particular, the positivity constraints); $\tilde{m}_1(\theta) = a_0 + a_1 \tilde{L}_0$ where $\tilde{L}_0$ is a fixed initial value (for instance 0) and $\tilde{m}_t(\theta) = m_t(\theta) = a_0 + a_1 L_{t-1}$ pour $t \geq 1$; initial values $\tilde{m}_0(\theta)$ and $\tilde{\sigma}_0^2(\theta)$, then

$$\tilde{\sigma}_t^2(\theta) = \omega + \alpha \{L_{t-1} - \tilde{m}_{t-1}(\theta)\}^2 + \beta \tilde{\sigma}_{t-1}^2(\theta), \quad t = 1, 2, \dots, n.$$

If $\eta_t \stackrel{iid}{\sim} \mathcal{N}(0, 1)$ then the QML is the ML (up to the initial values), but the QML is CAN under much less restrictive assumptions.

Estimation of VaR when $L_t = m_t(\theta) + \sigma_t(\theta)\eta_t$

- a) If the law F is assumed to be known for instance $F = \Phi$, then (for $t \geq n + 1$)

$$\widehat{\text{VaR}}_{t,1}(\alpha) = \tilde{m}_t(\hat{\theta}) + \tilde{\sigma}_t(\hat{\theta})\Phi^{-1}(1 - \alpha).$$

- b) If a parametric form is assumed for F , for instance a standardized St_ν , then $\theta = (a_0, a_1, \omega, \alpha, \beta, \nu)$ can be estimated by ML

$$\widehat{\text{VaR}}_{t,1}(\alpha) = \tilde{m}_t(\hat{\theta}_{MV}) + \tilde{\sigma}_t(\hat{\theta}_{MV})F_{\hat{\theta}_{MV}}^{-1}(1 - \alpha).$$

- c) Nonparametrically, one can estimate $F^-(1 - \alpha)$ by the $(1 - \alpha)$ -empirical quantile of the residuals

$$\hat{\eta}_s = \frac{L_s - \tilde{m}_s(\hat{\theta})}{\tilde{\sigma}_s(\hat{\theta})}, \quad s = 1, \dots, n$$

and

$$\widehat{\text{VaR}}_{t,1}(\alpha) = \tilde{m}_t(\hat{\theta}) + \tilde{\sigma}_t(\hat{\theta})F_n^-(1 - \alpha).$$

Estimation of VaR with $L_t = m_t(\theta) + \sigma_t(\theta)\eta_t$ (cont.)

d) The **Extreme Value Theory** can be used to estimate $F^-(1 - \alpha)$ (in particular for α very small).

Theorem (Extreme Value Theory)

For a very large class of laws P_X^a ,

$$X - u \mid X > u \sim GPD(\gamma, \sigma_u)$$

approximately when u is large, where

$$P\{GPD(\gamma, \sigma) \leq x\} = \begin{cases} 1 - \left(1 + \frac{\gamma x}{\sigma}\right)^{-1/\gamma}, & \gamma \neq 0 \\ 1 - e^{-x/\sigma}, & \gamma = 0, \end{cases}$$

with $x \geq 0$ for $\gamma \geq 0$, and $0 \leq x \leq -\sigma/\gamma$ for $\gamma < 0$.

► Example

^aSee for example, Embrechts, P., Kluppelberg, C., Mikosch, T., 1997. Modelling Extremal Events for Insurance and Finance. Springer, Berlin

EVT-based estimation of VaR when $L_t = m_t(\theta) + \sigma_t(\theta)\eta_t$

For a threshold u let N the number of residuals $\hat{\eta}_s > u$. By ML on $\{\hat{\eta}_s - u : \hat{\eta}_s > u\}$, one can estimated the tail parameter γ and the scale parameter σ .

Using the fact that, for $x > u$,

$$P(X > x) = P(X > u)P(X - u > x - u \mid X > u),$$

let (for $x > u$)

$$\hat{P}(\eta_t > x) = \frac{N}{n} \left(1 + \frac{\hat{\gamma}(x - u)}{\hat{\sigma}} \right)^{-1/\hat{\gamma}}.$$

For $\alpha < \frac{N}{n}$, one can take $\hat{F}^{-1}(1 - \alpha)$ such that

$$\frac{N}{n} \left(1 + \frac{\hat{\gamma}(\hat{F}^{-1}(1 - \alpha) - u)}{\hat{\sigma}} \right)^{-1/\hat{\gamma}} = \alpha, \text{ hence}$$

$$\hat{F}^{-1}(1 - \alpha) = u + \frac{\hat{\sigma}}{\hat{\gamma}} \left\{ \left(\frac{\alpha n}{N} \right)^{-\hat{\gamma}} - 1 \right\}.$$

EVT-based estimation of VaR (cont.)

To get asymptotic results, one have to assume that $u = u_n \rightarrow \infty$ and $N = N_n \rightarrow \infty$ as $n \rightarrow \infty$.

The most important practical problem is the choice of u (see McNeil, A.J. and Frey, R. (2000) Estimation of tail-related risk measures for heteroscedastic financial time series: an extreme value approach. *Journal of Empirical Finance* 7, 271-300).

Quantile Regression

Recall that the **linear regression** of a random variable Y on the k components of a random vector $\mathbf{X}$ is defined by

$$Y = \mathbf{X}'\boldsymbol{\beta} + \epsilon, \quad E(\epsilon \mid \mathbf{X}) = 0.$$

One can interpret $\mathbf{X}'\boldsymbol{\beta}$ as the optimal prediction of Y given the components of $\mathbf{X}$, in the mean square sense. The error term is constrained to have a null conditional expectation.

For $\alpha \in (0, 1)$, the **quantile regression** is defined by

$$Y = \mathbf{X}'\boldsymbol{\beta}_\alpha + \epsilon_\alpha, \quad F_{\epsilon_\alpha}^-(\alpha \mid \mathbf{X}) = 0.$$

If $X = 1$ then $\beta_\alpha = F_Y^-(\alpha)$.

The cost of an error $e = Y - \mathbf{X}'\boldsymbol{\beta}$ is $e^+\alpha + e^-(1 - \alpha)$ for the quantile regression, while it is e^2 for the standard linear regression.

Example of the linear scale-location model

Let

$$Y = \mathbf{X}'\boldsymbol{\theta} + (\mathbf{X}'\boldsymbol{\gamma})\epsilon,$$

where $\epsilon \sim F$ is independent of $\mathbf{X}$. Then

$$E(Y \mid \mathbf{X}) = \mathbf{X}'(\boldsymbol{\theta} + \boldsymbol{\gamma}E\epsilon)$$

and

$$F_Y^-(\alpha \mid \mathbf{X}) = \begin{cases} \mathbf{X}'\boldsymbol{\theta} + \mathbf{X}'\boldsymbol{\gamma}F_\epsilon^-(\alpha) & \text{if } \mathbf{X}'\boldsymbol{\gamma} \geq 0 \\ \mathbf{X}'\boldsymbol{\theta} - \mathbf{X}'\boldsymbol{\gamma}F_\epsilon^{-1}(1 - \alpha) & \text{if } \mathbf{X}'\boldsymbol{\gamma} < 0 \text{ and } F \text{ invertible} \end{cases}$$

Remark

To simulate Y it suffices to take $Y = \mathbf{X}'\boldsymbol{\theta} + \mathbf{X}'\boldsymbol{\gamma}F_\epsilon^-(U)$ with $U \sim \mathcal{U}_{[0,1]}$. More generally, to simulate a variable Y with quantile regression $\mathbf{X}'\boldsymbol{\beta}_\alpha \forall \alpha \in (0, 1)$ increasing in α , it suffices to set $Y = \mathbf{X}'\boldsymbol{\beta}_U$ ($F_{g(U)}^-(\alpha) = g(\alpha)$ when g is continuous and strictly increasing).

Estimation

Recall that

$$F_Z^-(\alpha) = \inf \left\{ \arg \min_u E \rho_\alpha(Z - u) \right\},$$

hence, the α -quantile of $Y - \mathbf{X}'\boldsymbol{\beta}_\alpha$ is null entails

$$0 = \inf \left\{ \arg \min_u E \rho_\alpha(Y - \mathbf{X}'\boldsymbol{\beta}_\alpha - u) \right\},$$

or

$$E \rho_\alpha(Y - \mathbf{X}'\boldsymbol{\beta}_\alpha) \leq E \rho_\alpha(Y - \mathbf{X}'\boldsymbol{\beta}), \quad \forall \boldsymbol{\beta}.$$

This justifies the estimator

$$\hat{\boldsymbol{\beta}}_\alpha = \arg \min_{\boldsymbol{\beta} \in \mathbb{R}^k} \sum_{i=1}^n \rho_\alpha(Y_i - \mathbf{X}_i'\boldsymbol{\beta}),$$

► Illustration

based on the sample $\begin{pmatrix} Y_1 \\ \mathbf{X}_1 \end{pmatrix}, \dots, \begin{pmatrix} Y_n \\ \mathbf{X}_n \end{pmatrix}$ of $\begin{pmatrix} Y \\ \mathbf{X} \end{pmatrix}$.

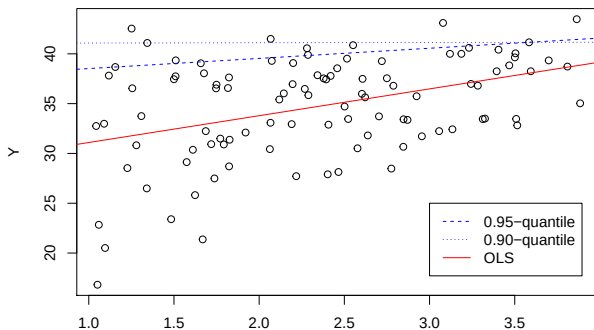
Incoherence problem, "Quantile crossing"

The estimators are not always coherent (monotonous):

$$\alpha_1 < \alpha_2 \not\Rightarrow \mathbf{X}'_i \hat{\boldsymbol{\beta}}_{\alpha_1} \leq \mathbf{X}'_i \hat{\boldsymbol{\beta}}_{\alpha_2} \quad \forall i,$$

whereas (under the strong linear model) we must have

$$\alpha_1 < \alpha_2 \Rightarrow \mathbf{X}'_i \boldsymbol{\beta}_{\alpha_1} \leq \mathbf{X}'_i \boldsymbol{\beta}_{\alpha_2} \quad \forall i.$$



QAR (Quantile Autoregressive) model

Koenker and Xiao (2006) proposed the **QAR(p)** model

$$Y_t = a_0(U_t) + a_1(U_t)Y_{t-1} + \cdots + a_p(U_t)Y_{t-p},$$

where (U_t) iid $\mathcal{U}_{[0,1]}$ and $a_i : [0, 1] \rightarrow \mathbb{R}$. If the function $u \rightarrow a_0(u) + a_1(u)Y_{t-1} + \cdots + a_p(u)Y_{t-p}$ is increasing, which is satisfied when $Y_{t-i} \geq 0$ and the a_i are quantile functions, we have

$$F_{Y_t}^{-1}(\alpha \mid Y_{t-1}, \dots, Y_{t-p}) = a_0(\alpha) + a_1(\alpha)Y_{t-1} + \cdots + a_p(\alpha)Y_{t-p}.$$

Estimation: For fixed α , one gets an estimator of $a(\alpha) = (a_0(\alpha), \dots, a_p(\alpha))'$ as solution of

$$\hat{a}(\alpha) = \arg \min_{a \in \mathbb{R}^{p+1}} \frac{1}{n} \sum_{i=1}^n \rho_{\alpha}(Y_t - a_0 - a_1 Y_{t-1} - \cdots - a_p Y_{t-p}).$$

CAViaR (Conditional Autoregressive VaR) model

Engle and Manganelli (2004) proposed to introduce **past quantiles** in the specification. Setting $q_t(\alpha) = F_{Y_t}^-(\alpha \mid Y_u, u < t)$, the model writes

$$q_t(\alpha) = a_0(\alpha) + \sum_{i=1}^p a_i(\alpha) g(Y_{t-i}) + \sum_{j=1}^r b_j q_{t-j}(\alpha)$$

where $g : \mathbb{R} \rightarrow [0, +\infty)$ (par exemple $g(y) = y^2$ ou $g(y) = |y|$), with increasing functions a_i and $b_j \geq 0$ (to get q_t increasing).

Examples: For α fixed,

$$q_t = a_0 + a|Y_{t-1}| + bq_{t-1}$$

or the GARCH(1,1) CAViaR

$$q_t = \sqrt{\omega + aY_{t-1}^2 + bq_{t-1}^2}$$

with $\omega, a, b > 0$.

DAQ (Dynamic Additive Quantile) model

Gouriéroux and Jasiak (2007) considered quantile functions $a_i(\cdot)$, $i = 1, \dots, p$ with the same support as Y_t .³ Then $\forall c_1, \dots, c_p \geq 0$, $\forall c_0 \in \mathbb{R}$ if $Y_t(\Omega) = \mathbb{R}$, $c_0 = 0$ if $Y_t(\Omega) = [0, \infty)$, the quantile function $c_0 + \sum_{i=1}^p c_i a_i(\cdot)$ has the same support as Y_t . The model is thus

$$q_t(\alpha) = c_0(Y_u, u < t) + \sum_{i=1}^p c_i(Y_u, u < t) a_i(\alpha),$$

which has the advantage of increasing with α . For instance, one can take

$$\begin{aligned} q_t(\alpha) = & d_0 + d_1 |Y_{t-1} - \mu| + \sqrt{\omega + a Y_{t-1}^2} \Phi^{-1}(\alpha) \\ & + \sqrt{\omega^* + a^* Y_{t-1}^2} \tan(\pi(\alpha - 1/2)). \end{aligned}$$

³ $a_i(u) \xrightarrow{u \rightarrow 0} -\infty$ and $a_i(u) \xrightarrow{u \rightarrow 1} +\infty$ if $Y_t(\Omega) = \mathbb{R}$; $a_i(u) \xrightarrow{u \rightarrow 0} 0$ and $a_i(u) \xrightarrow{u \rightarrow 1} +\infty$ if $Y_t(\Omega) = [0, \infty)$

- 1 Quantile estimation
- 2 Estimation of risk measures
- 3 Backtesting and numerical illustrations
 - Backtesting
 - Empirical comparison of VaR estimations
 - Few recent works

Principle of the Backtesting

The idea is to check if the observed ex-post losses are in accordance with the risk measures computed ex-ante.

Consider the VaR at horizon 1.

"In-sample" data are used to calibrate the estimation method for the VaR (choice of a model, estimation of the volatility parameters and of the residuals quantile, for instance).

"Out-of-sample" data allow to compare the violations of VaR with the nominal level α . The comparison must be **model free** (backtesting does not depend on the VaR prediction method).

Hit variables

Prudential regulations require an assessment of the VaR (and the other risk measures) by Backtesting procedures which are "model free". Let the "hit" variable

$$I_{t+1}(\alpha) = 1_{L_{t,t+1} > \text{VaR}_{t,1}(\alpha)}.$$

If $\text{VaR}_{t,1}(\alpha)$ is really a VaR at level α , we have

$$P(L_{t,t+1} > \text{VaR}_{t,1}(\alpha)) = \alpha, \quad I_t \sim \mathcal{B}(\alpha).$$

If $\text{VaR}_{t,1}(\alpha)$ is a conditional VaR at level α , we have

$P_t(L_{t,t+1} > \text{VaR}_{t,1}(\alpha)) = \alpha$ and, for $k > 0$,

$$EI_t I_{t+k} = E(E_{t+k-1} I_t I_{t+k}) = E(I_t E_{t+k-1} I_{t+k}) = \alpha E I_t = \alpha^2.$$

Thus $\text{Cov}(I_t, I_{t+k}) = 0$, which is equivalent to I_t and I_{t+k} independent. Finally, we have

$$(I_t) \text{ iid } \sim \mathcal{B}(\alpha).$$

Tests

Christoffersen, P. F. (1998) proposed tests for the hypotheses

$$H_0^{\text{UC}} : P(I_t = 1) = \alpha, \quad H_0^{\text{Ind}} : P(I_t = 1 \mid I_{t-1} = 0) = P(I_t = 1 \mid I_{t-1} = 1)$$

of unconditional coverage (UC) and independence (Ind). These tests are Likelihood Ratio (LR) tests based on n out-of-sample observations $I_1, \dots, I_n$. The maximum likelihood is

$$L(I_1, \dots, I_n, \hat{\alpha}) = \hat{\alpha}^{T_1} (1 - \hat{\alpha})^{n-T_1} \text{ with } T_1 = \sum_{t=1}^n I_t \text{ and } \hat{\alpha} = T_1/n. \text{ We have}$$

$$\text{LR}_{\text{UC}} = 2 \log \frac{\hat{\alpha}^{T_1} (1 - \hat{\alpha})^{n-T_1}}{\alpha^{T_1} (1 - \alpha)^{n-T_1}}$$

which follows asymptotically a χ_1^2 under H_0^{UC} . Hence the critical region $\{\text{LR}_{\text{UC}} > \chi_1^2(1 - \underline{\alpha})\}$ at level $\underline{\alpha}$ (proposed by Kupiec, 1995).

Testing the independence of violations

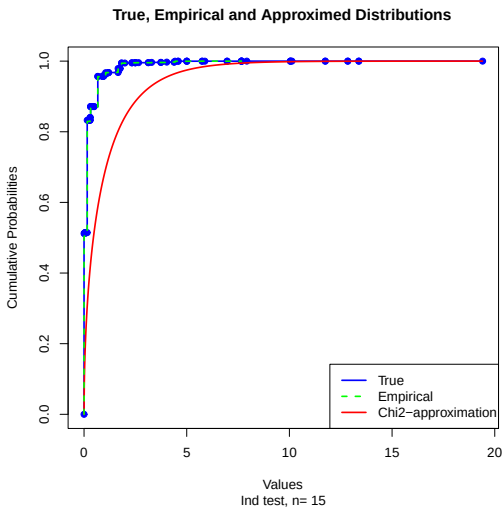
$$H_0^{\text{Ind}} : P(I_t = 1 \mid I_{t-1} = 0) = P(I_t = 1 \mid I_{t-1} = 1).$$

For a Markov chain with transition probabilities p_{01} and p_{11} , the maximum likelihood has the form $L(I_1, \dots, I_t, \hat{p}_{01}, \hat{p}_{11})$; under the constraint $p_{01} = p_{11}$ it is given by $L(I_1, \dots, I_t, \hat{p}_{11}^c)$. The LR test statistic is thus

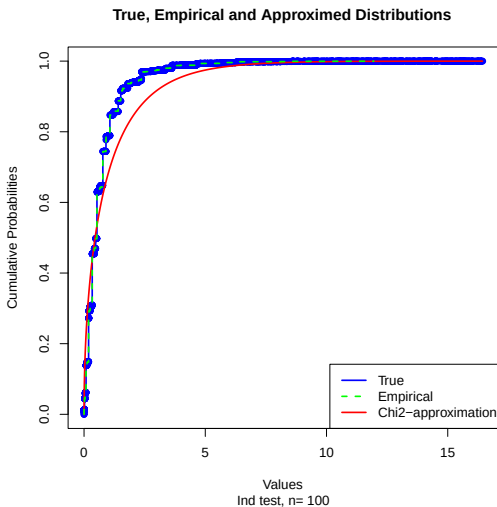
$$\text{LR}_{\text{Ind}} = 2 \log \frac{L(I_1, \dots, I_t, \hat{p}_{01}, \hat{p}_{11})}{L(I_1, \dots, I_t, \hat{p}_{11}^c)}$$

which follows asymptotically a χ_1^2 distribution under H_0^{Ind} . Hence the critical region $\{\text{LR}_{\text{Ind}} > \chi_1^2(1 - \underline{\alpha})\}$ at level $\underline{\alpha}$.

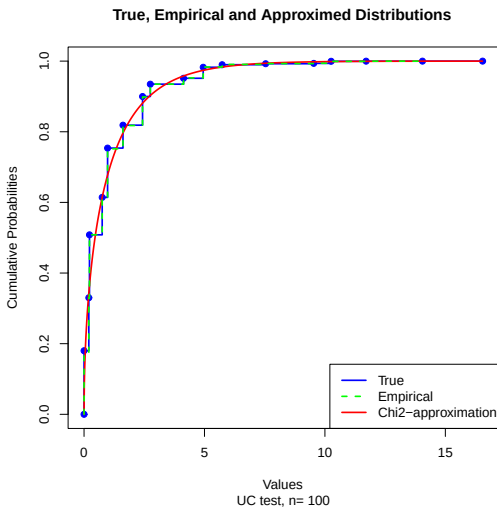
The chi-squared approximation is very bad for n small



Still inaccurate for $n = 100$



Also for the UC backtest



Using fast dynamic-programming algorithms

the new R package [ExactVaRTest](#) created by **Yujian Chen** solves the problem.

[Try a R code using this package here.](#)

The chi-squared approximations are very inaccurate

$$H_0^{\text{Ind}} : P(I_t = 1 \mid I_{t-1} = 0) = P(I_t = 1 \mid I_{t-1} = 1).$$

For a Markov chain with transition probabilities p_{01} and p_{11} , the maximum likelihood has the form $L(I_1, \dots, I_t, \hat{p}_{01}, \hat{p}_{11})$; under the constraint $p_{01} = p_{11}$ it is given by $L(I_1, \dots, I_t, \hat{p}_{11}^c)$. The LR test statistic is thus

$$\text{LR}_{\text{Ind}} = 2 \log \frac{L(I_1, \dots, I_t, \hat{p}_{01}, \hat{p}_{11})}{L(I_1, \dots, I_t, \hat{p}_{11}^c)}$$

which follows asymptotically a χ_1^2 distribution under H_0^{Ind} . Hence the critical region $\{\text{LR}_{\text{Ind}} > \chi_1^2(1 - \underline{\alpha})\}$ at level $\underline{\alpha}$.

The chi-squared approximations are very inaccurate, but the new **R package ExactVaRTest by Yujian Chen** solves the problem.

Empirical comparison of several VaR estimators

Kuester, Mittnik and Paoletta (2006)⁴ backtest univariate estimation methods on 30 years of daily returns of NASDAQ.

- The parameters are estimated on rolling windows of size 1000;
- Backtests on 6681 estimates of the VaR at horizon 1;
- The hypothesis of iid violations is rejected (Table 2) for the marginal VaR estimators (HS, Normal, t , EVT);
- CAViAR-type methods perform badly (Table 5);
- AR(1)-GARCH(1,1) with heavy tail distribution for the noise, or the semi-parametric methods (EVT, FHS) perform best (Table 3).

⁴Kuester, K., Mittnik, S., and Paoletta, M. S. (2006) [Value-at-risk prediction: A comparison of alternative strategies](#). *Journal of Financial Econometrics* 4, 53–89.

Limitations of the study

- A method which is not rejected by any backtest is not necessarily good⁵;
- The rationality of the ST-EVT method is questionable;;
- No Monte Carlo experiment (on real data, the true DGP (if existing) is unknown);
- This type of study must be redone by others, on other series, with other methods ... (but painful to do and difficult to value);
- No multivariate method to exploit interactions between risk factors;⁶

► A small Monte Carlo experiment

⁵Independently of the data, setting $\text{VaR} = +\infty$ with probability $1 - \alpha$ and $\text{VaR} = -\infty$ with probability α results in iid violations with probability α .

⁶Santos, A. A., Nogales, F. J., and Ruiz, E. (2012) Comparing univariate and multivariate models to forecast portfolio value-at-risk. *Journal of financial econometrics* 11, 400–441.

Elicitability

Definition A risk measure $r : \mathcal{M} \rightarrow \mathbb{R}$ is elicitable if there exists a loss/cost/scoring function $s(x, y) \rightarrow [0, \infty)$ such that

$$r(L) = \arg \min_x E s(x, L), \quad \forall L \in \mathcal{M}.$$

Example If $\mathcal{M}$ is a set of continuous losses, $\text{VaR}(\alpha)$ is elicitable

$$\text{VaR}_L(\alpha) = \arg \min_x E \rho_{1-\alpha}(L - x).$$

Appli r_1 is preferred to r_2 if (out-of-sample)

$$\frac{1}{n} \sum_{i=1}^n s(r_1(L_i), L_i) << \frac{1}{n} \sum_{i=1}^n s(r_2(L_i), L_i) \quad (\text{Diebold-Mariano test}).$$

See Exam 2016

Backtesting ES^{9 10}

- ES is unobservable (VaR also);
- Contrary to VaR, ES is not elicitable;⁷
- Jointly (VaR, ES) is elicitable⁸, but that seems difficult to use.

For $L_t \mid \mathcal{F}_{t-1} \sim F_{t-1}$ continuous strictly increasing, let $U_t = F_{t-1}(L_t) \sim U_{[0,1]}$ and the cumulative violation process

$$H_t(\alpha) := \frac{1}{\alpha} \int_0^\alpha 1_{L_t > \text{VaR}_t(u)} du = \frac{1}{\alpha} \int_0^\alpha 1_{U_t > 1-u} du.$$

⁷Gneiting, T. (2011) [Making and evaluating point forecasts](#). *Journal of the American Statistical Association* 106, 746–762.

⁸Fissler, T., and Ziegel, J. F. (2016) [Higher order elicibility and Osband's principle](#). *The Annals of Statistics* 44, 1680–1707.

⁹Du, Z. and Escanciano, J. C. (2017) [Backtesting expected shortfall: accounting for tail risk](#). *Management Science* 63, 901–1269.

¹⁰Couperier, O. and Leymarie, J. (2018) [Backtesting expected shortfall via multi-quantile regression](#).

Backtesting ES

Du and Escanciano (2017) noted that

$$H_t(\alpha) = \frac{1}{\alpha} \int_{1-\alpha}^1 1_{U_t > v} dv = 1_{U_t > 1-\alpha} \frac{U_t - 1 + \alpha}{\alpha}$$

have mean $\alpha/2$ and variance $\alpha/3 - \alpha^2/4$. Moreover $\{H_t(\alpha) - \alpha/2\}_t$ is a martingale difference. If the [assumed model](#) gives a good approximation $\hat{U}_t = \hat{F}_{t-1}(L_t)$ of U_t , the statistics

$$\frac{1}{\sqrt{n}} \sum_{t=1}^n \frac{\hat{H}_t(\alpha) - \frac{\alpha}{2}}{\sqrt{\alpha(\frac{1}{3} - \frac{\alpha}{4})}}, \quad \hat{H}_t(\alpha) = 1_{\hat{U}_t > 1-\alpha} \frac{\hat{U}_t - 1 + \alpha}{\alpha},$$

should be approximatively $\mathcal{N}(0, 1)$ distributed.

Systemic Risk Measures

The main aim is to assess the contribution of a given financial institution to the overall risk of an entire financial sector, to better identify the vulnerabilities of the financial system.

- Marginal expected shortfall (MES);
- Systemic expected shortfall (SES);
- Systemic risk measure (SRISK);
- ΔCoVaR ;
- capital shortfall computed from the regulatory banking stress tests;
- systemic risk scoring to identify systemically important financial institutions (SIFIs).

Marginal expected shortfall (MES)^{11 12}

Let Y_{1t} the stock return of a given financial institution and Y_{2t} the market return. Assume that the distribution of $Y_t = (Y_{1t}, Y_{2t})$ is continuous and integrable.

At the risk level α and given some information set $\mathcal{F}_{t-1}$,

$$\text{MES}_{1t}(\alpha) = E_{t-1}(-Y_{1t} \mid Y_{2t} < -\text{VaR}_{2t}(\alpha)),$$

where $\text{VaR}_{2t}(\alpha)$ is the market VaR conditional to $\mathcal{F}_{t-1}$. This is the expected loss of the financial institution given that the loss of the market is larger than its $\text{VaR}(\alpha)$. This can be estimated by fitting a parametric MGARCH on (Y_t) .

¹¹Acharya, V., Pedersen, L., Philippon, T., and Richardson, M. (2017) Measuring systemic risk. *Review of Financial Studies* 30, 2–47.

¹²Brownlees, C. and Engle, R. (2017). Srisk: A conditional capital shortfall measure of systemic risk. *Review of Financial Studies* 30, 48–79.

CoVaR^{13 14 15}

At the risk levels α and β , the CoVaR of the financial firm is $\text{CoVaR}_{1t}(\beta, \alpha)$ such that

$$\beta = P_{t-1}(Y_{1t} < -\text{CoVaR}_{1t}(\beta, \alpha) \mid Y_{2t} < -\text{VaR}_{2t}(\alpha)).$$

Note that

$$\text{MES}_{1t}(\alpha) = -E\left(F_{Y_{1t}|Y_{2t} < -\text{VaR}_{2t}(\alpha); \mathcal{F}_{t-1}}^{-1}(U)\right) = \int_0^1 \text{CoVaR}_{1t}(u, \alpha) du.$$

¹³Adrian, T. and Brunnermeier, M. (2016) [CoVaR](#). *American Economic Review* 106, 1705–1741.

¹⁴Banulescu, D., Hurlin, C., Leymarie, J., and Scaillet, O. (2021) [Backtesting Marginal Expected Shortfall and Related Systemic Risk Measures](#). *Swiss Finance Institute Research Paper* 19-48.

¹⁵Francq, C., and Zakoïan, J. M. (2025) [Inference on dynamic systemic risk measures](#). *Journal of Econometrics* 247, 105936.

ΔCoVaR^{16}

Let $\alpha < \underline{\alpha} < \bar{\alpha}$. The median CoVaR, $\text{CoVaR}_{1t}(\alpha, \underline{\alpha}, \bar{\alpha})$, is the CoVaR conditional on a median state for the market:

$$\alpha = P_{t-1}(Y_{1t} < -\text{CoVaR}_{1t}(\alpha, \underline{\alpha}, \bar{\alpha}) \mid -\text{VaR}_{2t}(\underline{\alpha}) < Y_{2t} < -\text{VaR}_{2t}(\bar{\alpha})).$$

The stressed CoVaR, $\text{CoVaR}_{1t}(\alpha)$, is the CoVaR conditional on a distress state for the market:

$$\alpha = P_{t-1}(Y_{1t} < -\text{CoVaR}_{1t}(\alpha) \mid Y_{2t} < -\text{VaR}_{2t}(\alpha)).$$

The ΔCoVar is

$$\Delta\text{CoVar}_{1t}(\alpha) = \text{CoVaR}_{1t}(\alpha) - \text{CoVaR}_{1t}(\alpha, \underline{\alpha}, \bar{\alpha}).$$

¹⁶Adrian, T. and Brunnermeier, M. (2016) [CoVaR](#). *American Economic Review* 106, 1705–1741.

On the inference of dynamic risk measures

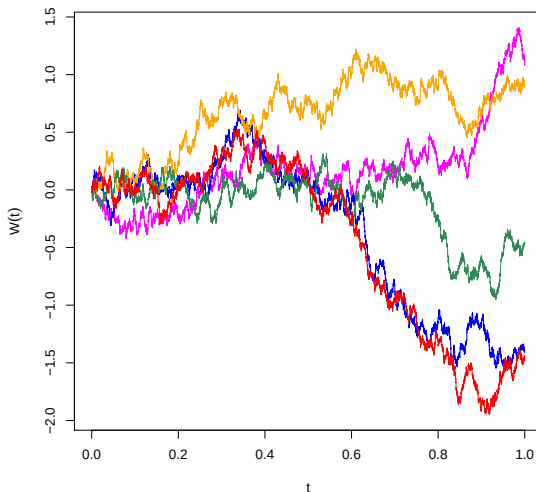
- [Christoffersen and Gonçalves \(2005\)](#), and [Spierdijk \(2014\)](#): resampling techniques to account for parameter estimation uncertainty in univariate dynamic models;
- [Escanciano and Olmo \(2010, 2011\)](#): robust backtesting procedures in presence of estimation risk (and also of model risk);
- [Gouriéroux and Zakoïan \(2013\)](#): asymptotic bias in the coverage probabilities induced by estimation, and corrected VaR;
- [Francq and Zakoïan \(2015, 2018, 2020\)](#): asymptotic confidence intervals for the conditional VaR in semi-parametric uni and multivariate frameworks;
- [Hurlin, Laurent, Quaedvlieg and Smeekes \(2017\)](#): bootstrap-based comparison tests of two conditional risk measures;
- [Hienemann \(2019\)](#) [Bootstrap inference for conditional risk measures](#)

End of Chapter 3 😊 ! End of the course 😊 !

Simulation of a Brownian motion

5 simulations d'un mouvement Brownien

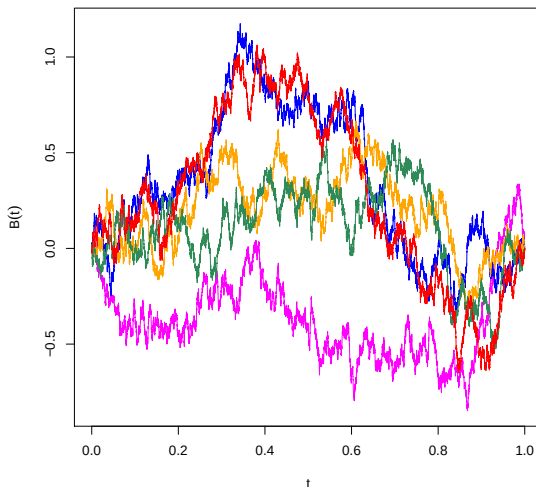
◀ Definition



Simulation of a Brownian bridge

5 simulations d'un pont Brownien

◀ Definition



Equivalence of the two definitions (already shown in a more general framework)

Note that

- $\varphi_{n,\alpha}(z) = (\alpha - 1)(\bar{X}_n - z)$ for $z > X_{n,n}$ and $\varphi_{n,\alpha}(z) = \alpha(\bar{X}_n - z)$ for $z < X_{1,n}$;
- $\varphi_{n,\alpha}$ convex and $\varphi_{n,\alpha}(z) \rightarrow \infty$ when $|z| \rightarrow \infty$;
- strictly decreasing before $\xi_{n,\alpha}$ and increasing after $\xi_{n,\alpha}$.

We thus have the characterisation

$$\frac{\partial \varphi_{n,\alpha}(z)}{\partial z^-} < 0 \text{ for } z < \xi_{n,\alpha} \quad \text{and} \quad \frac{\partial \varphi_{n,\alpha}(z)}{\partial z^+} \geq 0 \text{ for } z \geq \xi_{n,\alpha}.$$

Equivalence of the two definitions (continued)

We have

$$\frac{\partial \rho_\alpha(x-z)}{\partial z^-} = -\alpha + 1_{z > x} \quad \text{and} \quad \frac{\partial \rho_\alpha(x-z)}{\partial z^+} = -\alpha + 1_{z \geq x},$$

since

$$\rho_\alpha(x-z) = \begin{cases} (x-z)(\alpha-1) & \text{if } z \in [x, \infty) \text{ (derivative } 1-\alpha) \\ (x-z)\alpha & \text{if } z \in (-\infty, x] \text{ (derivative } -\alpha) \end{cases}$$

We thus have

[◀ Return](#)

$$\frac{1}{n} \sum_{i=1}^n 1_{X_i < z} < \alpha \text{ for } z < \xi_{n,\alpha} \quad \text{and} \quad \frac{1}{n} \sum_{i=1}^n 1_{X_i \leq z} \geq \alpha \text{ for } z \geq \xi_{n,\alpha}$$

which shows that $\xi_{n,\alpha}$ is indeed the smallest $X_{k,n}$ such that $\frac{k}{n} \geq \alpha$.

Example

If $X \sim \text{GPD}(\gamma, \sigma)$ then

$$X - u \mid X > u \sim \text{GPD}(\gamma, \sigma + \gamma u)$$

since

$$\begin{aligned} P(X - u \leq y \mid X > u) &= \frac{P(u < X \leq y + u)}{P(X > u)} \\ &= \frac{\left(1 + \frac{\gamma u}{\sigma}\right)^{-1/\gamma} - \left(1 + \frac{\gamma(y+u)}{\sigma}\right)^{-1/\gamma}}{\left(1 + \frac{\gamma u}{\sigma}\right)^{-1/\gamma}} \\ &= 1 - \left(1 + \frac{\gamma y}{\sigma + \gamma u}\right)^{-1/\gamma} \end{aligned}$$

Interpretation of Lindeberg's condition

Under the condition, the contribution of any individual variable Y_{ni} to the sum becomes small as $n \rightarrow \infty$, in the sense that

$$\sup_{1 \leq i \leq n} \frac{\sigma_{ni}^2}{s_n^2} \rightarrow 0.$$

Indeed

$$\begin{aligned} \sup_{1 \leq i \leq n} \frac{\sigma_{ni}^2}{s_n^2} &\leq \sup_{1 \leq i \leq n} \frac{EY_{ni}^2 1_{\{|Y_{ni}| < \varepsilon s_n\}}}{s_n^2} + \frac{1}{s_n^2} \sum_{i=1}^n EY_{ni}^2 1_{\{|Y_{ni}| \geq \varepsilon s_n\}} \\ &\leq \varepsilon^2 + \frac{1}{s_n^2} \sum_{i=1}^n EY_{ni}^2 1_{\{|Y_{ni}| \geq \varepsilon s_n\}}. \end{aligned}$$

Lindeberg's CLT contains the usual CLT

Let $Z_1, \dots, Z_n$ an iid sequence with finite variance,

$$Y_{ni} = \frac{Z_i - EZ_i}{\sqrt{n}}.$$

We have $\sigma_{ni}^2 = n^{-1}\text{Var}(Z_0)$ and $s_n^2 = \text{Var}(Z_0)$. Moreover

$$\begin{aligned} \sum_{i=1}^n EY_{ni}^2 1_{\{|Y_{ni}| \geq \varepsilon\}} &= \sum_{i=1}^n n^{-1} \int_{\{|Z_i - EZ_i| \geq \sqrt{n}\varepsilon\}} |Z_i - EZ_i|^2 dP \\ &= \int_{\{|Z_1 - EZ_1| \geq \sqrt{n}\varepsilon\}} |Z_1 - EZ_1|^2 dP \rightarrow 0 \end{aligned}$$

because $\{|Z_1 - EZ_1| \geq \sqrt{n}\varepsilon\} \downarrow \emptyset$ and $\int_{\Omega} |Z_1 - EZ_1|^2 dP < \infty$.
 Lindeberg's condition is thus satisfied.

◀ Return

Intuition about the QMLE

Under some regularity conditions, the ergodic theorem entails

$$\hat{\theta} \rightarrow \arg \min_{\theta \in \Theta} E \left\{ \log \sigma_t^2(\theta) + \frac{(L_t - m_t(\theta))^2}{\sigma_t^2(\theta)} \right\}.$$

Moreover

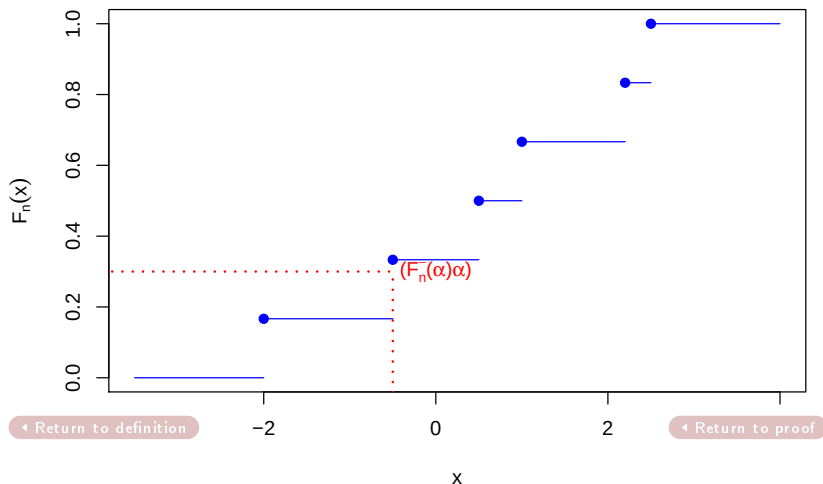
$$E(\epsilon_t^2 \mid \mathcal{F}_{t-1}) = \arg \min_{\sigma_t^2 \in \mathcal{F}_{t-1}} E \left\{ \log \sigma_t^2 + \frac{\epsilon_t^2}{\sigma_t^2} \right\}.$$

Indeed the function $\varphi(\sigma^2) = E(\epsilon^2/\sigma^2 + \log \sigma^2)$ is minimized at $\sigma^2 = E\epsilon^2$:

$$\varphi(E\epsilon^2) - \varphi(\sigma^2) = 1 + \log(E\epsilon^2/\sigma^2) - E\epsilon^2/\sigma^2 \leq 0.$$

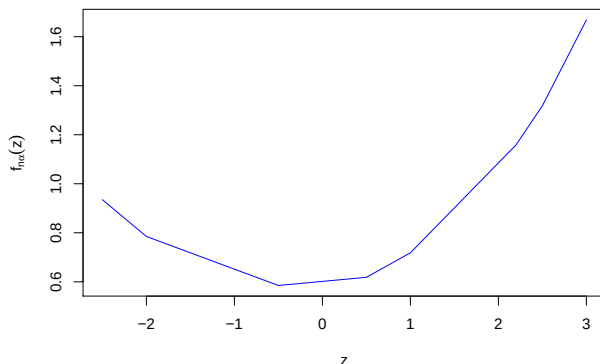
$$X_{1,6} = -2, X_{2,6} = -0.5, X_{3,6} = 0.5, X_{4,6} = 1, X_{5,6} = 2.2, X_{6,6} = 2.5;$$

$$\alpha = 0.3$$



$$X_{1,6} = -2, X_{2,6} = -0.5, X_{3,6} = 0.5, X_{4,6} = 1, X_{5,6} = 2.2, X_{6,6} = 2.5$$

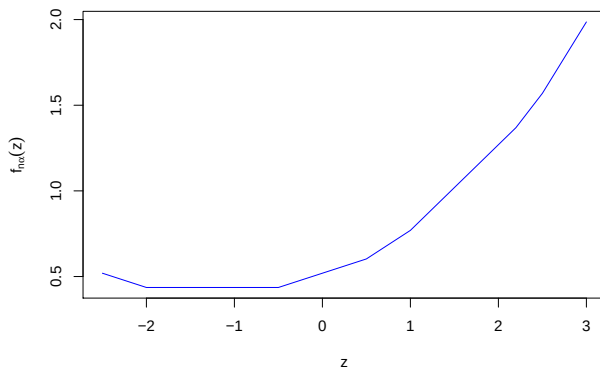
$$\alpha=0.3$$



◀ Return

$$X_{1,6} = -2, X_{2,6} = -0.5, X_{3,6} = 0.5, X_{4,6} = 1, X_{5,6} = 2.2, X_{6,6} = 2.5$$

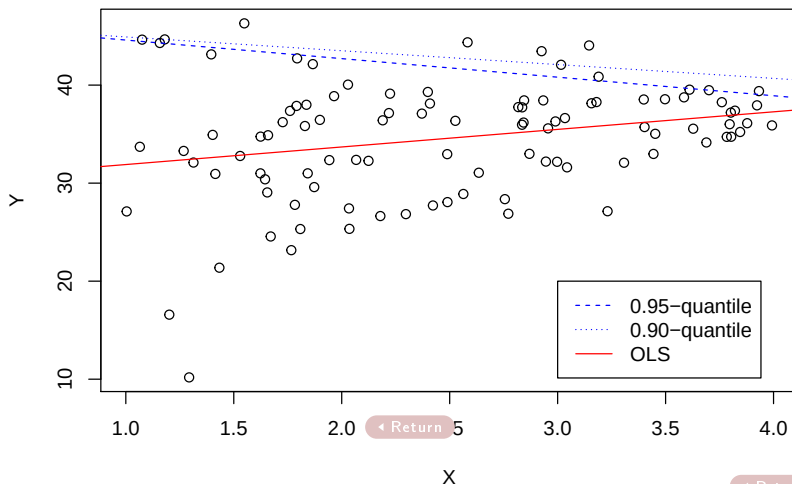
$$\alpha=1/6$$



◀ Return

Ordinary and quantile regressions (heterogenous effect of X)

$X \propto$ minimum wage; $Y \propto$ income



◀ Return

◀ Return

Data Generating Process (DGP)

Let a portfolio of value $V_t = \sum_{i=1}^d a_i p_{t,i}$, where the variations of the risk factors $\log p_{t,i}$ are denoted $r_{t,i} = \log p_{t,i} / p_{t-1,i}$.

DGP: The vector $r_t = (r_{t,1}, \dots, r_{t,d})'$ follows a CCC-GARCH(1,1) model defined by

$$r_t = H_t^{1/2} \eta_t, \quad H_t = D_t R D_t$$

where η_t is a strong white noise with standardized Student distribution, that is, $\eta_t = \sqrt{\frac{\nu-2}{\chi_\nu^2}} \mathcal{N}(0, I_d)$ for $\nu > 2$, R is a correlation matrix, D_t is a diagonal matrix of diagonal terms $\sqrt{h_{i,t}}$ such that

$$h_{t,i} = \omega_i + \alpha_i r_{t,i}^2 + \beta_i h_{t-1,i}.$$

The parameters $\omega_i > 0$, $\alpha_i > 0$ and $\beta_i \geq 0$ as well as ν are such that r_t is strictly stationary with conditional variance H_t .

Description of the simulation experiments

Independent replications ($\text{Iter} = 1, \dots, 7$) of simulations of length $n = 20\,000$.

The initial portfolio has value $V_0 = 3\,000$, equally diversified between the $d = 3$ assets: $a_i p_{0,i} = 1\,000$ pour $i = 1, 2, 3$.

The first $n_1 = 4\,000$ observations are used for estimating the volatility parameters. The remaining observations are reserved for backtesting.

The different parameter sets

In all the experiments, the noises η_t are simulated in the trivariate Student distribution with $\nu = 7$ degrees of freedom.

Jeu A The volatility parameters are close $10^4\omega = (0.04, 0.04, 0.04)$, $\alpha = (0.04, 0.03, 0.05)$, $\beta = (0.89, 0.9, 0.88)$ and the assets are highly correlated $R(1, 2) = -0.855$, $R(1, 3) = 0.855$ and $R(2, 3) = -0.81$.

Jeu B One very persistent asset, not the other two ones, $10^4\omega = (0.04, 0.04, 0.04)$, $\alpha = (0.04, 0.03, 0.15)$, $\beta = (0.95, 0.0, 0.0)$. The persistent asset is uncorrelated with the other 2 ones $R(1, 2) = 0$, $R(1, 3) = 0$ and $R(2, 3) = 0.9$.

Losses, conditional and unconditional univariate VaR

Suppose $V_t > 0$ and let $\epsilon_t = \log V_t/V_{t-1}$ the log-return of the portfolio. The loss at horizon 1 is

$$L_{t,t+1} = V_t - V_{t+1} = V_t(1 - e^{\epsilon_{t+1}})$$

and the VaR is

$$\text{VaR}_{t,1}(\alpha) = V_t(1 - e^{q_{t,1}(\alpha)}),$$

where $q_{t,1}(\alpha)$ is the α -quantile of the law of the future return ϵ_{t+1} (conditional on V_t for a non conditional VaR, or conditional on $\{V_u, u \leq t\}$ for a conditional univariate VaR, that is based on the past values of the portfolio).

Multivariate conditional VaR

To take advantage of the multivariate relations between r_t and its past values, note that

$$L_{t,t+1} = V_t - V_{t+1} = \sum_{i=1}^d a_i p_{t,i} - \sum_{i=1}^d a_i p_{t,i} e^{r_{t+1,i}} = \sum_{i=1}^d a_i p_{t,i} (1 - e^{r_{t+1,i}}).$$

By linearization we get

$$L_{t,t+1} \simeq - \sum_{i=1}^d a_i p_{t,i} r_{t+1,i} = -(a_1 p_{t,1}, \dots, a_d p_{t,d}) H_{t+1}^{1/2} \eta_{t+1}.$$

Because the law of η_t is spherical, we have

$$\begin{aligned} \text{VaR}_{t,1}(\alpha) &\simeq - \left\| (a_1 p_{t,1}, \dots, a_d p_{t,d}) H_{t+1}^{1/2} \right\| F_{\eta}^{-1}(\alpha) \\ &= \left\| (a_1 p_{t,1}, \dots, a_d p_{t,d}) H_{t+1}^{1/2} \right\| F_{|\eta|}^{-1}(1 - 2\alpha). \end{aligned}$$

Methods

- **HS** (for historical) denotes the method in which

$$\text{VaR}_{t,1}(\alpha) := F_{L_{t,t+1}}^-(1 - \alpha) = V_t F_{L_{t,t+1}/V_t}^-(1 - \alpha)$$

is estimated by V_t times the $(1 - \alpha)$ -empirical quantile of the last $K = 250$ relative losses $L_{t-s,t-s+1}/V_{t-s}$ for $s = 1, \dots, K$. The relative losses seem "more stationary" than the losses.

- **RM** (for RiskMetrics) estimates the VaR by approaching the quantile $q_{t,1}(\alpha)$ in

$$\text{VaR}_{t,1}(\alpha) = V_t(1 - e^{q_{t,1}(\alpha)})$$

by $\hat{\sigma}_{t+1} \Phi^{-1}(1 - \alpha)$ where $\hat{\sigma}_t$ is obtained recursively by $\hat{\sigma}_t^2 = 0.06\epsilon_{t-1}^2 + 0.94\hat{\sigma}_{t-1}^2$.

Methods (cont.)

- GARCH** estimates the quantile $q_{t,1}(\alpha)$ by $\hat{\sigma}_{t+1}\hat{F}_t^{-1}(1-\alpha)$. The volatility is obtained recursively by the GARCH equation $\hat{\sigma}_t^2 = \hat{\omega} + \hat{\alpha}\epsilon_{t-1}^2 + \hat{\beta}\hat{\sigma}_{t-1}^2$, where the parameters are estimated by QMV on the first n_1 values. The quantile $\hat{F}_t^{-1}(1-\alpha)$ is obtained from the residuals $\hat{\eta}_u = \epsilon_u/\hat{\sigma}_u$ for $u = r_0 + 1, \dots, t$ (with $r_0 = 10$ to alleviate the effect of the initial values).
- CCC** : using the trivariate returns $r_1, \dots, r_{n_1}$, the parameters of a CCC-GARCH(1,1) are estimated by QML. This allows to obtain an estimator $\hat{H}_t$ of the volatility and the residuals $\hat{\eta}_t = \hat{H}_t^{-1/2}\epsilon_t$. The VaR is estimated by its theoretical approximation, by replacing H_{t+1} by $\hat{H}_{t+1}$ and $F_{|\eta|}^{-1}(1-2\alpha)$ by the $(1-2\alpha)$ -empirical quantile of the absolute values of the residual components available at time t .

Backtesting for the set of parameters A and the level $\alpha = 5\%$

Iter	Violations (in %)				average relative VaR (in %)			
	HS	RM	GARCH	CCC	HS	RM	GARCH	CCC
1	5.5	5.0	6.0	4.9	0.536	0.549	0.501	0.551
2	5.4	5.1	5.2	4.9	0.606	0.618	0.595	0.620
3	5.4	5.2	6.5	5.1	0.683	0.703	0.637	0.689
4	5.5	5.3	5.7	4.8	0.809	0.829	0.791	0.832
5	5.5	4.9	5.1	4.3	0.567	0.591	0.570	0.603
6	5.4	5.4	5.6	5.1	0.903	0.906	0.886	0.908
7	5.5	5.3	5.9	5.3	0.571	0.583	0.542	0.564

Iter	p -value of the test LR_{uc}				p -value of the test LR_{ind} (en %)			
	HS	RM	GARCH	CCC	HS	RM	GARCH	CCC
1	0.81	88.33	0.00	51.35	0.05	51.21	52.89	1.05
2	2.38	61.13	18.21	40.30	0.01	6.26	0.84	15.70
3	1.10	26.29	0.00	44.70	72.92	11.94	21.73	3.64
4	0.99	7.80	0.01	20.17	18.71	48.17	72.84	60.27
5	0.42	63.77	58.62	0.01	0.18	66.62	68.68	60.54
6	3.13	3.13	0.04	66.28	0.03	27.89	1.74	11.25
7	0.53	9.78	0.00	14.94	17.63	32.66	53.94	70.93

Backtesting for the set of parameters A and the level $\alpha = 1\%$

Iter	Violations (in %)				average relative VaR (in %)			
	HS	RM	GARCH	CCC	HS	RM	GARCH	CCC
1	1.4	1.7	1.4	1.0	0.841	0.776	0.800	0.872
2	1.4	1.8	1.1	0.9	0.929	0.872	0.929	0.974
3	1.4	1.6	1.3	0.9	1.033	0.992	1.005	1.074
4	1.4	1.8	1.2	0.9	1.249	1.170	1.274	1.318
5	1.5	1.6	1.0	0.8	0.896	0.834	0.922	0.955
6	1.4	1.7	1.1	0.9	1.373	1.279	1.412	1.439
7	1.5	1.6	1.2	1.0	0.887	0.824	0.874	0.893

Iter	p -value of the test LR_{uc}				p -value of the test LR_{ind} (en %)			
	HS	RM	GARCH	CCC	HS	RM	GARCH	CCC
1	0.00	0.00	0.00	52.20	37.93	34.53	60.36	64.36
2	0.00	0.00	7.39	14.50	60.36	21.66	23.17	53.06
3	0.00	0.00	0.00	19.62	42.15	22.13	56.87	55.28
4	0.00	0.00	1.37	14.50	13.68	60.99	32.19	53.06
5	0.00	0.00	58.02	1.74	20.25	68.95	5.97	13.99
6	0.00	0.00	24.00	6.11	0.23	72.30	5.91	3.82
7	0.00	0.00	0.72	75.02	75.54	0.30	2.79	63.41

Backtesting for the set of parameters B and the level $\alpha = 1\%$

Iter	Violations (in %)				average relative VaR (in %)			
	HS	RM	GARCH	CCC	HS	RM	GARCH	CCC
1	1.6	1.7	0.9	1.0	1.368	1.301	1.443	1.423
2	1.3	1.8	0.7	1.0	0.378	0.349	0.428	0.381
3	1.5	1.8	0.7	1.0	0.562	0.526	0.624	0.585
4	1.6	1.7	1.1	0.7	3.090	2.936	3.223	3.421
5	1.5	1.7	1.0	0.9	3.940	3.597	4.013	4.150
6	1.4	1.7	0.2	0.9	0.407	0.378	0.575	0.417
7	1.6	1.6	1.0	0.9	1.889	1.718	1.899	1.943

Iter	p -value of the test LR_{uc}				p -value of the test LR_{ind} (en %)			
	HS	RM	GARCH	CCC	HS	RM	GARCH	CCC
1	0.00	0.00	29.53	75.10	61.36	38.42	58.66	55.43
2	0.00	0.00	0.02	99.94	24.84	84.89	19.50	73.50
3	0.00	0.00	0.06	52.20	0.27	51.11	17.99	67.57
4	0.00	0.00	13.83	0.06	6.59	16.77	42.00	86.38
5	0.00	0.00	58.02	7.36	0.63	15.09	52.57	11.99
6	0.00	0.00	0.00	16.92	4.05	83.93	69.94	18.77
7	0.00	0.00	99.94	10.46	32.67	83.77	31.34	50.86

Results

- Results vary much from simulation to the other. This can be explained by the fact that the series V_t is non stationary, that the different paths are not similar.
- Table show that the frequencies of violations are far from the nominal values for the methods HS and RM (except for RM in the case $\alpha = 5\%$, due to the fact that quantile of the standardized Student is close to that of the Gaussian for $\alpha = 5\%$). Such methods thus have poor performance.

Results (cont.)

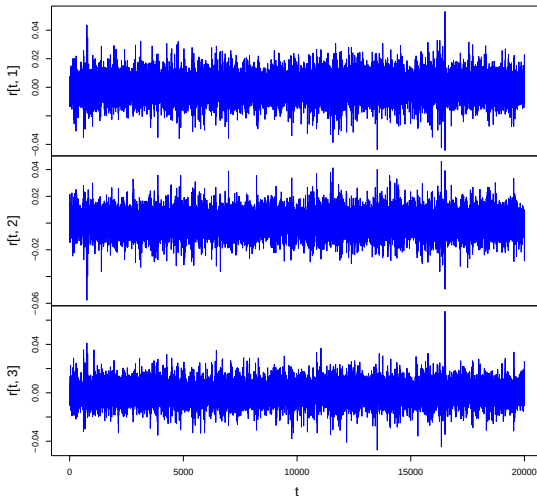
- The GARCH and CCC methods are very similar for the set A. From a regulation point of view, the CCC method is preferable because the frequency of rejection is often below the expected one. For the set B, the CCC method is superior, with correct frequencies of violations of the VaR correctes and much smaller average VaR. This is not surprising as the univariate GARCH is applied to an aggregate series, and thus cannot account for the fact that the 3 risk factors have very different dynamics.

◀ Return

10 simulated trajectories of the value (V_t) of the portfolio

Simulation of the 3 risk factors of the portfolio

◀ Return



Loss and estimated VaR on a short period

Return

